CCS6334
DIGITAL & COMPUTER FORENSICS

Lecture 2: Hardware & Software Environments

1

Learning Objectives

After studying this chapter, you should be able to explain:

- The variety of computing and storage devices and the nature of digital information they hold of potential evidentiary value
- OS software and applications used in the creation, transfer, and storage of digital information
- Identifying and explaining filesystems and files that contain evidence of value and where they are typically stored
- Password security and encryption impacts on protecting information and concealing evidence

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The Nature of Digital Information

- Digital evidence comprises digital information found on a broad range of electronic devices
- Any information held in digital data format that is useful to the forensic examiner because of its value in various legal proceedings
- Sources of potential evidence stored on digital devices include e-mails, audio and video files, electronic documents, spreadsheets, databases, system logs, and filesystem data

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Handling Computer Hardware

- You should develop an understanding of computer hardware for a number of reasons.
- Certain types of systems and hardware support only certain types of software for operating systems, file systems, and applications.
- For example, it is important to understand that an Intel-based Mac can support both macOS and its related APFS and HFS+ file systems. Nevertheless, that same computer can also support a Windows operating system and related NTFS file system when Boot Camp is running.
- Being cognizant of the diversity of computer hardware is also necessary because you need to know how systems can be connected to external devices, such as routers or external hard drives.
- These connected devices, such as routers, often contain digital evidence and may need to be seized if a warrant permits.
- The investigator might also need to reconstruct the computer and its devices when she returns to the laboratory.

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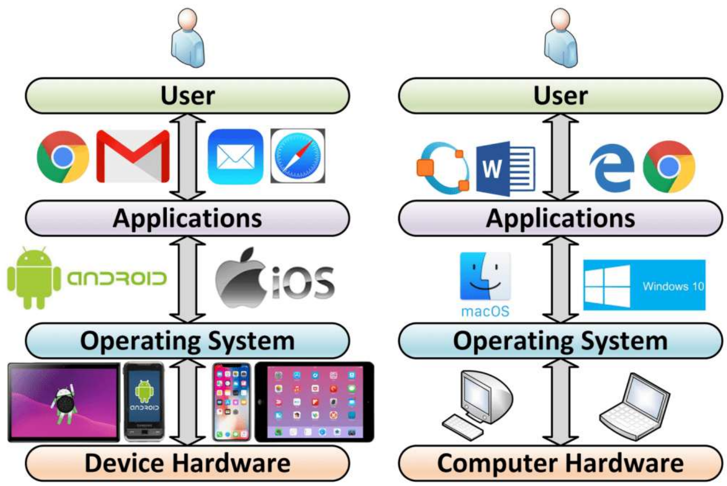
Layer of Computing System



- Evidence is generated from each layer
- User activities= Applications+ OS+ Disk/Memory

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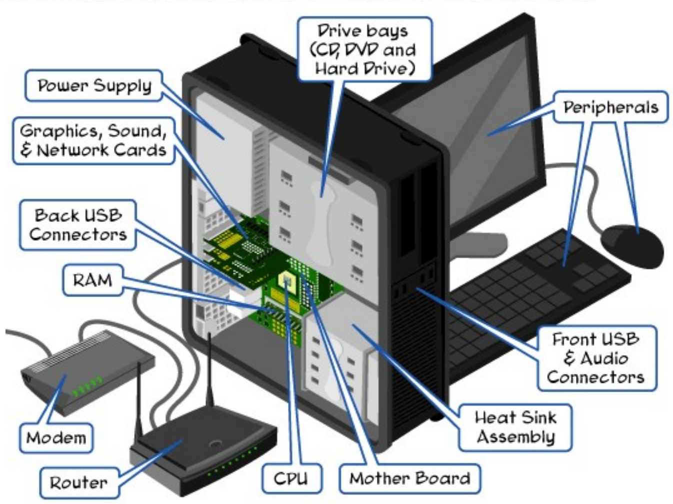
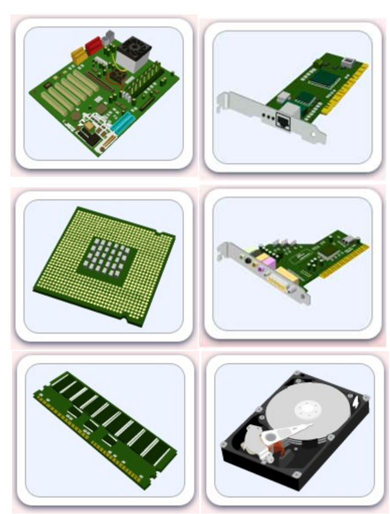
Parts of Computing System





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Parts of Computing System

```
C:\Users\Zu[redacted]>systeminfo

Host Name:                 [redacted]ZUL[redacted]
OS Name:                   Microsoft Windows 11 Education
OS Version:                10.0.26200 N/A Build 26200
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          User
Registered Organization:   N/A
Product ID:                00378-60470-05129-AA582
Original Install Date:     26/12/2024, 8:14:22 PM
System Boot Time:          5/11/2025, 2:27:02 PM
System Manufacturer:       Dell Inc.
System Model:              Latitude 3450
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 186 Steppi
ng 3 GenuineIntel ~1300 Mhz
BIOS Version:              Dell Inc. 1.16.1, 15/8/2025
Windows Directory:         C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume3
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+08:00) Kuala Lumpur, Singapore
Total Physical Memory:     24,261 MB
```

```
Virtualization-based security: Status: Running
                               Required Security Properties:
                               Available Security Properties:
                                    Base Virtualization Suppor
                                    Secure Boot
                                    DMA Protection
                                    UEFI Code Readonly
                                    SMM Security Mitigations 1
                                    Mode Based Execution Contr
                                    APIC Virtualization
                               Services Configured:
                               Services Running:
                               App Control for Business policy:
                               App Control for Business user mo
                               Security Features Enabled:
Hyper-V Requirements:          A hypervisor has been detected.
```

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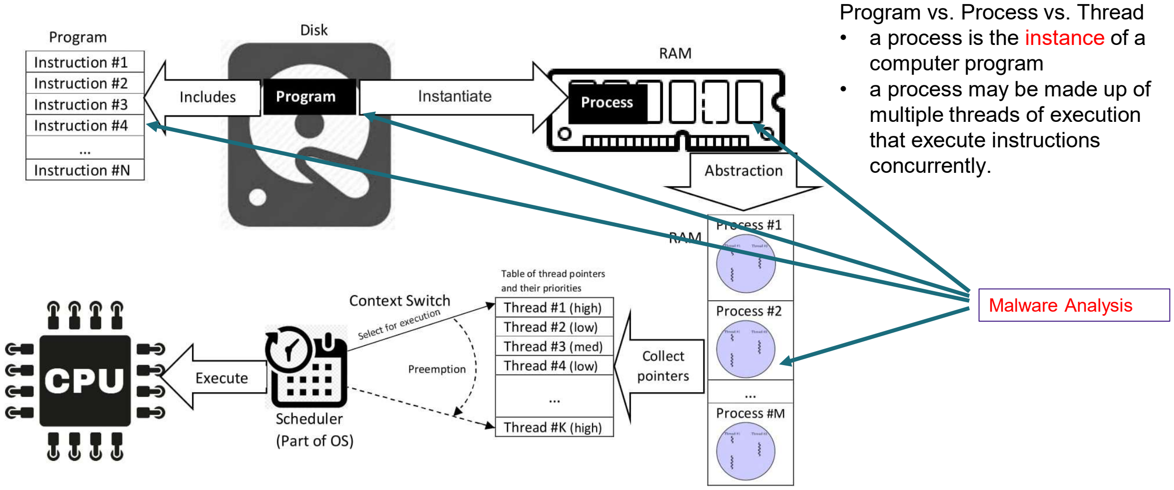
How Computer Work?



Program vs. Process vs. Thread

- a process is the instance of a computer program
- a process may be made up of multiple threads of execution that execute instructions concurrently.

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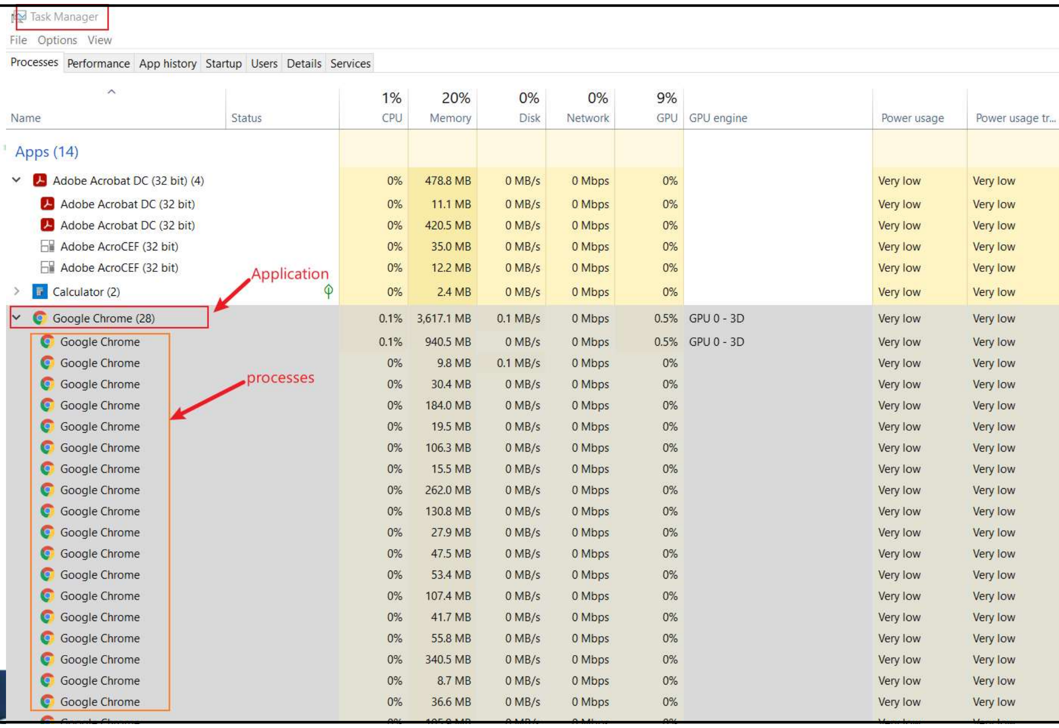

Name	Status	1% CPU	20% Memory	0% Disk	0% Network	9% GPU	GPU engine	Power usage	Power usage tr...
Apps (14)									
Adobe Acrobat DC (32 bit) (4)		0%	478.8 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Adobe Acrobat DC (32 bit)		0%	11.1 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Adobe Acrobat DC (32 bit)		0%	420.5 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Adobe AcroCEF (32 bit)		0%	35.0 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Adobe AcroCEF (32 bit)		0%	12.2 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Calculator (2)		0%	2.4 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome (28)		0.1%	3,617.1 MB	0.1 MB/s	0 Mbps	0.5%	GPU 0 - 3D	Very low	Very low
Google Chrome		0.1%	940.5 MB	0 MB/s	0 Mbps	0.5%	GPU 0 - 3D	Very low	Very low
Google Chrome		0%	9.8 MB	0.1 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	30.4 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	184.0 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	19.5 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	106.3 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	15.5 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	262.0 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	130.8 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	27.9 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	47.5 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	53.4 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	107.4 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	41.7 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	55.8 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	340.5 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	8.7 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Google Chrome		0%	36.6 MB	0 MB/s	0 Mbps	0%		Very low	Very low

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Why computer forensics is hard?

- **Technical difficulties**
  - OS: different OS types, versions, complexity of OS
  - Applications: many applications, different version of applications
  - Hardware: CPU, GPU, camera
  - Ever-advancing technology: e.g., database changes, SSD vs HDD, EV car
- **A complex and connected world**
  - IoT devices: Alex, Camera, Fitbit, Smart Phone
- **Information explosion**
  - too much information, different type of evidence
  - how to collect, analyze, validate them systematically?
- **Computer forensics is NOT only a computer science discipline**
  - criminal justice, law, computer science, security

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DIGITAL STORAGE

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Magnetic Storage

- Primary means of storage and still remains popular in the form of disks and tape → you've covered this in operating systems
- Magnetic tapes and disks store digital data in binary form, as ones and zeroes, with the use of a electromagnetic device (read/write head) – data is saved permanently without the need for constant power (i.e. non-volatile)
- The simple and consistent design of these devices has been advantageous to forensics practitioners using well-established processes such as dead analysis and forensic tools such as write blockers to recover evidence.
- Advanced tools such as scanning electron microscopes can also possibly recover data from magnetic media (see MFM, SFM, Lorentz microscopy and Bitter technique)
- Is being gradually replaced by solid state storage but for extreme quantities and long-term storage of data, is still the media of choice due to cost/byte

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Optical Storage

- CDs, DVDs, and Blu-ray discs are optical storage media that are relatively inexpensive and are capable of read-only bulk storage
- The data embedded on these media is also non-volatile and, generally, what has been written to the media is recoverable in theory
- Although (relatively) easily disposed of, microscopy techniques have been used to manually recover data from damaged optical storage but even this method has its limits.

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Digital Storage - SSD

- Solid-state storage (sometimes interchanged with the term flash storage) are fast emerging as the storage media of choice due to its speed and low power requirements
- Although it functions similarly to a magnetic disk drive, solid-state memory works very differently compared to traditional methods when used in a hard disk.
- The way solid-state memory operates tend to make forensic tools not work correctly when used for analysis → new tools required
- Built-in operations like wear-levelling (TRIM), onboard encryption, NAND over-provisioning make extraction far more difficult.

Solid State Drive (SSD)

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Digital Storage – SSD Caveats

- With all the benefits of SSD, also comes several caveats (i.e. limitations)
- Traditional methods of recovery are not as effective on SSDs
- Deletions are instantaneous and often are unretrievable
- Increasing numbers of device manufacturers are now having flash storage permanently attached and cannot be easily accessed → failure of the main device results in inaccessible storage

  → Examples = phone storage, modern laptops, smart devices
- SSDs also suffer from premature failure that is often final and instantaneous
- Traditional magnetic storage may fail slowly thus affording last minute recovery methods

Solid State Drive (SSD)

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Handling Disk Copy

- Two processes are used by computer forensics examiners for making a bit-for-bit copy of a hard drive:
- A **disk clone** is an exact copy of a hard drive and can be used as a backup for a hard drive because it is bootable just like the original.
- A **disk image** is a file or a group of files that contain bit-for-bit copies of a hard drive but cannot be used for booting a computer or other operations.
- The process of cloning a hard drive is a faster process than imaging a hard drive. The time difference between the two processes is substantial. Therefore, when a computer forensics examiner works undercover or perhaps needs to obtain a copy of a hard drive and leaves the computer with the custodian, cloning the drive is more practical.

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Network-stored Data

- Data need not be stored locally – options for NAS, storage servers etc
- ✓ SAN = Storage Area Networks / NAS = Network Attached Storage
- ✓ Access to the data is provided by connecting a device to the network and undergoing authentication methods
- For practitioners this may require an entirely different approach to performing forensics operations]
- Often times it is desirable to make physical images of entire systems but network-stored data presents new sets of issues → size, server downtime, live (overwritten) data etc

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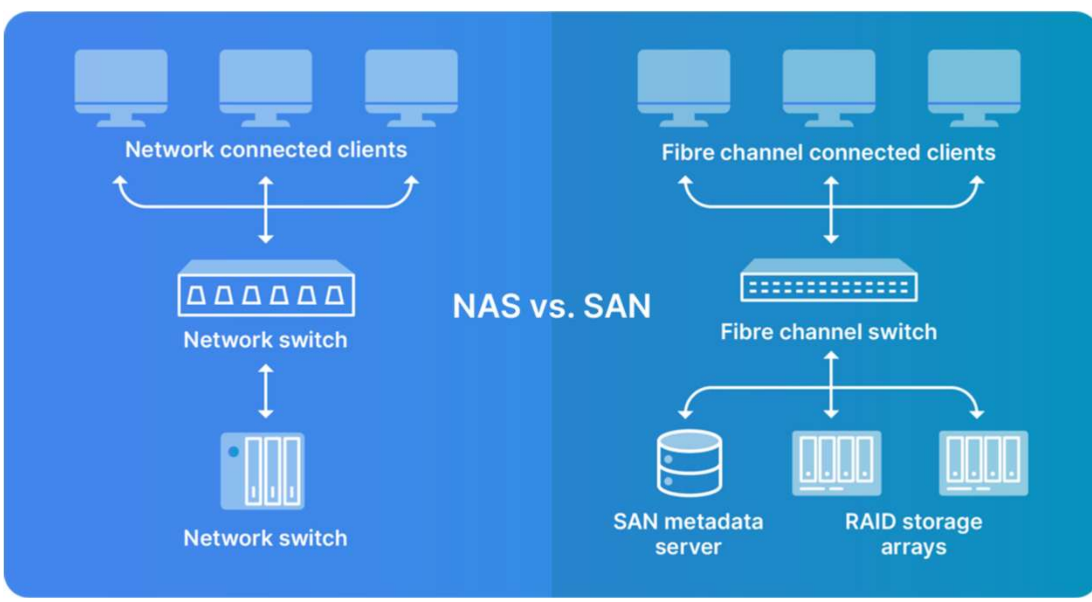

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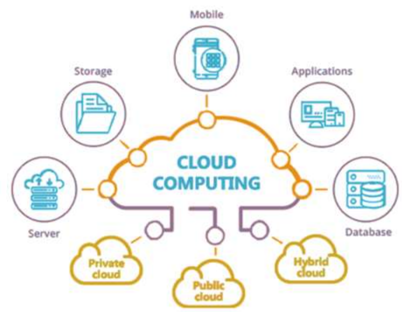
Cloud-based Services & Storage

- Internet-based computing more commonly referred to as *the cloud* that shares resources and information has become a popular network feature used by individuals, organizations, and government agencies
  - From storage to computing resources, many different services can be provided on the cloud
- The issue here is finding evidence in larger datasets dispersed over networks is proving problematic for conventional recovery tools
- One of the challenges in recovering data is difficulty in identifying the path that the data takes from the cloud server to the storage device, which may not be a fixed route
  - Additionally the server may be located in one or more locations in different legal jurisdictions



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Cloud-based Services & Storage

- Finding where the data is stored on the cloud is already an issue → Additionally, the server may be located in one or more locations in different legal jurisdictions.
  → The ease to make multiple copies makes distinguishing files difficult
- Increasing implementation of E2E encryption without hosting knowledge makes accessibility difficult when applying brute force methods
- When a service goes down, loss of data is often catastrophic
  - E.g. Megaupload server closure resulted in loss of over 25,000TB of data equivalent to approx. 12 billion+ unique files from >100 million users

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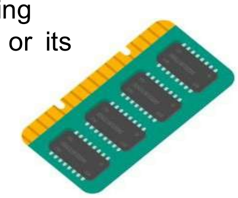
Random Access Memory

- A form of computer storage that can be read and quickly changed in any order – typically used to store running data and executing machine codes.
- Volatile form of memory of which the storage is not meant for long-term because of the requirement for constant refreshing and continual power supply.
- Capturing information stored in RAM may be important as it provides details of the most recent use of the device and includes some keyboard activity and running execution variables (e.g. encryption keys)
- Difficult to do as processes can quickly overwrite areas in RAM and powering down the device may make it impossible to regain access to the device or its contents



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OPERATING SYSTEM & SOFTWARE

Mac

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Operating System & Software

- A set of programs that control access between devices (e.g. I/O devices, storage, network, etc) and application software programs
- Various OSes exist – Windows, Linux, MacOS, Android, PalmOS, iOS – that cover different devices and hardware platforms
- Applications interact with OS functions to provide operations such as file saves, file transfers, copies and so on → files leave records on the device which can possibly be recovered as evidence
- Devices (internal/external) interact via drivers – small programs that provide communication capabilities → possible intercept points

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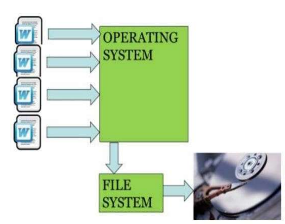
Filesystems and its Categories

- The way file information is stored varies among different operating systems and storage devices → FAT, NTFS, ExFAT, EXT, ISO, Joliet, HSF, etc.
- Commands received from the operating system in order to read and write files are interpreted in a directory structure, incorporating a file index system that defines file naming protocols and the maximum size of the file → This has been covered before in Operating Systems course



Writing files to a hard drive

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Filesystems and its Categories (Cont..)

- The filesystem records the general file information which is a unique structure on each individual device
- The filesystem is also rich in file metadata (you have learnt about metadata in database systems, but this is similar but not the same!)
  - To practitioners, metadata forms an essential part to the navigation and examination of filesystem information
  - Assists greatly in reconstructing events of relevance to a case
- However, data loss/corruption makes additional analysis more difficult and reliance on older (possibly invalid) backup copies of data and records is required
- On a Windows-based machine (majority of businesses use these but the same logic applies for other OSes) the filesystem consists of the filename, metadata and content categories

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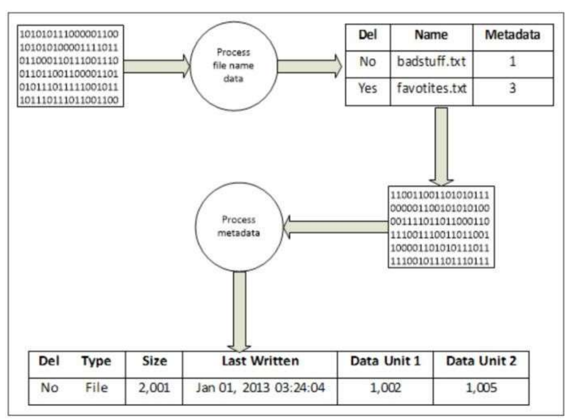
Filesystems and its Categories (Cont..)

- The filename category catalogs (i.e. keeps track) of data used to assign a name to each file
  - in the form of directories, subdirectories, files, etc. each as a metadata address for each file
- The operating system stores all file data and metadata in binary form, which is translated to human-readable text or images through the GUI



Filename information schema

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Filesystems and its Categories (Cont..)

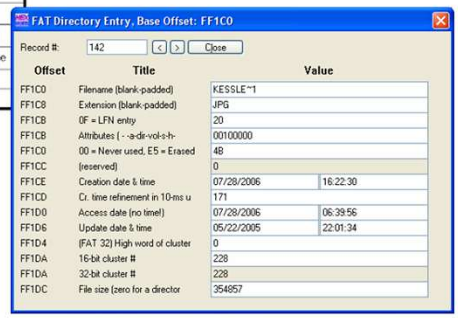
Root Directory SFN Entry Data Structure

Bytes	Purpose
0	First character of file name (ASCII) *or* allocation status (0x00=unallocated, 0xe5=deleted)
1-10	Characters 2-11 of the file name (ASCII); the "." is implied between bytes 7 and 8
11	File attributes (see File Attributes table)
12	Reserved
13	File creation time (in tenths of seconds)*
14-15	Creation time (hours, minutes, seconds)*
16-17	Creation date*
18-19	Access date*
20-21	High-order 2 bytes of address of first cluster (0 for FAT12/16)*
22-23	Modified time (hours, minutes, seconds)
24-25	Modified date
26-27	Low-order 2 bytes of address of first cluster
28-31	File size (0 for directories)

File Attributes

Flag Value	Description
0000 0001 (0x01)	Read-only
0000 0010 (0x02)	Hidden file
0000 0100 (0x04)	System file
0000 1000 (0x08)	Volume label
0000 1111 (0x0f)	Long file name
0001 0000 (0x10)	Directory
0010 0000 (0x20)	Archive

** Bytes 13-22 are unused by DOS*



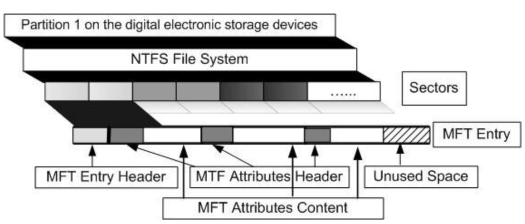
FAT Directory Entry, Base Offset: FF1C0

Record #: 142

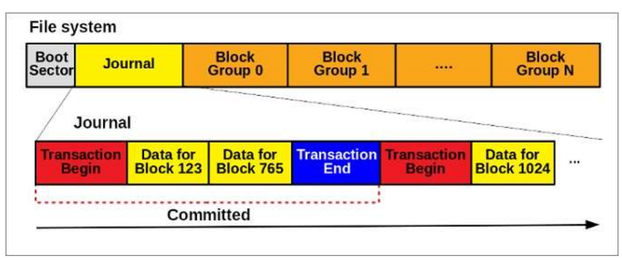
Offset	Title	Value	
FF1C0	Filename (blank-padded)	KESSLE~1	
FF1C8	Extension (blank-padded)	JPG	
FF1CB	0F = LFN entry	20	
FF1CB	Attributes (- -a-dir-vol-s-h-	00100000	
FF1C0	00 = Never used, E5 = Erased	4B	
FF1CC	(reserved)	0	
FF1CE	Creation date & time	07/28/2006	16:22:30
FF1CD	Cr. time refinement in 10-ms u	171	
FF1D0	Access date (no time!)	07/28/2006	06:39:56
FF1D6	Update date & time	05/22/2005	22:01:34
FF1D4	(FAT 32) High word of cluster	0	
FF1DA	16-bit cluster #	228	
FF1DA	32-bit cluster #	228	
FF1DC	File size (zero for a director	354857	

FAT filesystem data structure

Filesystems and its Categories (Cont..)



NTFS (Windows)



EXT3 (Linux)

Metadata

- Deleted filenames and their corresponding metadata addresses are often used to recover file content
- Being able to use filename listings is a fundamental part of forensic examinations as it allows the practitioner to identify the names of the files and parent directories and can be used for searching for evidence based on filename, path, or file extension.
- However, if the metadata address is cleared during file deletion, it may not be possible to locate further information. If only part of a filename is known, it is still possible to search using that part
- When a file is deleted, the metadata entry is changed to the unallocated state, and the operating system may wipe some of the file values
- Some consumer-grade file recovery software operate using this method to retrieve deleted files

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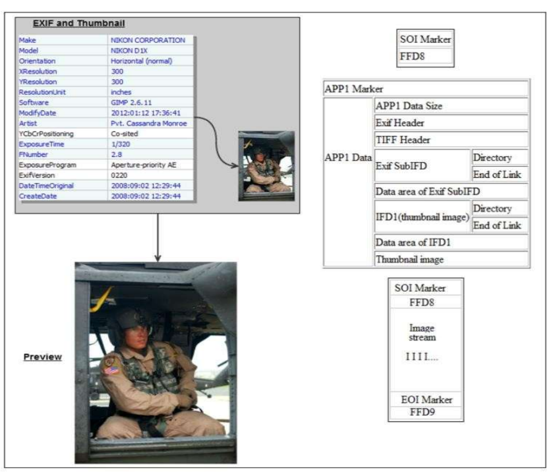
Metadata - Example



Metadata of a thumbnail file of a JPEG image

- The figure shows metadata recovered from a thumbnail of a photographic image generated by photo apps
- Even after the original file is deleted, this small database file may remain, containing miniature versions of the original file and file metadata.
- In this example, digital photo metadata known as Exchangeable Image File (EXIF) is created by the device
- Can provide information such as the location of the image taken, date, time, and serial number of the camera

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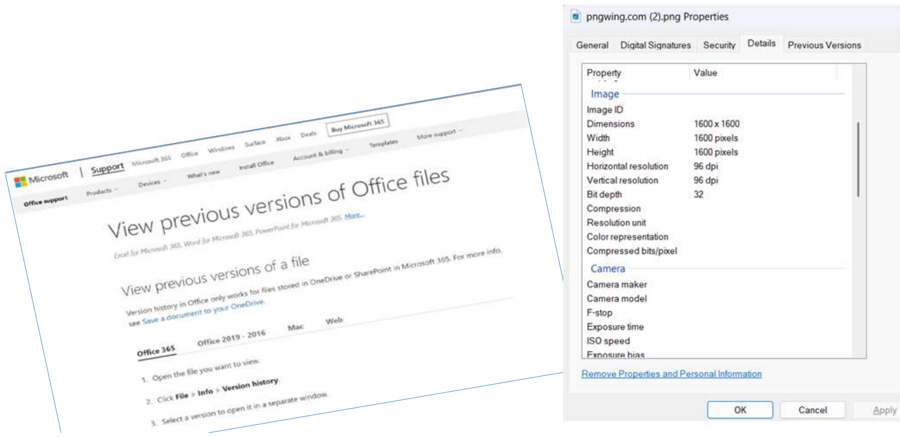
Metadata - Example

- Different digital files carry different metadata – some even older versions of the file itself!



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Metadata - Example

- Metadata becomes a double-edged sword
  - can be beneficial but also detrimental if not treated properly
- For example, no forensic protection was used to protect file integrity, simply copying of the file from one location to another on the same computer will automatically alter the metadata → considered as evidence contamination and can have serious implications for a legal case because it invites challenge from opposing team
- Thus, the importance of protecting digital evidence and highlights various processes and forensic tools used to prevent contamination of the evidence is required

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Content Category

- The content category consists of the actual contents of a file, such as the text written to a document file, figures added to a spreadsheet, or a picture inside an image file
- If recovered from unallocated space, it may have no linked metadata or filename, and the only clues to its antecedents may be gleaned from the file signature and clues garnered from the contents, especially text documents.
- File content can also be faked, altered, and/or manipulated alongside the filename and metadata to throw off forensics' practitioners from a digital trail.

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Locating Evidence in Filesystems

- The nature of any transgression to some extent dictates the type of relevant evidence that can possibly be recovered
- One of the fundamental challenges practitioners face is determining with any certainty the link between a suspect and the digital evidence recovered
- The practitioner only collects all relevant evidence that supports various hypotheses, but it is for others, such as juries, to decide whether the evidence helps determine guilt or innocence
- The practitioner must be guided by the evidence and if that proves inconclusive, he or she must look for more evidentiary clues to offer likely hypotheses as to what happened
- In a digital environment, we are looking for the digital "smoking gun"

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Locating Evidence in Filesystems (Cont.)

- The challenge to practitioners is locating information or data of relevance to the case under investigation
  - reason – some transgression, offence or illegal activity for recovering data (i.e. the WHAT of a scenario)
- This is often required as probable cause to get access to the device(s) involved in the event (i.e. the WHERE of the identified scenario)
- Once obtained, it is looking for associated evidence that correlates with and corroborates the key evidence (i.e. the WHO, WHEN, and HOW)
- It is also common during the examination; evidence of other transgressions may be recovered that could possibly be related
- In traditional forms of crime, investigators try to determine the means, opportunity, and motive for the transgressions being carried out

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The Means of Transgression

- Investigators look for the means or suitable process a suspect used to carry out an illegal act
- The use of an application installed on a recovered device and linked to the transgression may record activities and may be useful for demonstrating how the transgression occurred
  - E.g. the use of e-mail messaging to send a death threat or connecting to the Internet to download illegal pornography will leave an audit trail/ event logs to allow a reconstruction of what happened and the processes involved
  - Note that it is easy to imply an e-mail account recording its actions, it does not necessarily establish that it was sent from the device by a suspect → The process could have been completed by another person accessing the account remotely using a different computer!!

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The Means of Transgression (Cont.)

- The practitioner would have to determine where the truth lay and undertake a thorough analysis of the e-mail message in relation to the computer being examined
- Absolute concrete evidence to indict that a particular person (suspect) was indeed in possession of the source device at a particular point in time and performed the transgression is often needed in the court of law
- Another aspect of determining whether a suspect had the means to commission an offense is verifying whether the suspect had the computer skills to use the software involved, such as in the case of forging an electronic document or manipulating a photograph

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The Opportunity of Transgression

- Having the opportunity to do something illegal seems straightforward but proving a suspect alone had opportunity through access of the device may be problematic
- It may be difficult, if not impossible, to link the time of a crime to a suspect's access to the computer or network in the absence of any corroboration.
- Someone's access via logins can be seen via audit logs and assumptions made that the login is valid but if unauthorized access was gained to the account it may prove to be difficult to validate → unless some other evidence corroborates the record

- Audit and access user logs are not infallible and can be altered and falsified and are therefore not always reliable.
- Time and date stamps and file locations of key events might help confirm the circumstances relating to a transgression

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The Opportunity of Transgression (Cont.)

- Computer user access security may prohibit unauthorized access transgression and establish user identity
- ACLs might be a source to help narrow down list of suspects responsible for a particular transgression
- Ultimately, determining who really had access to the device/network has to be established → Often, this is inconclusive and it is unwise to assume the obvious In criminal cases, it must be proven beyond reasonable doubt, and to a lesser extent in civil cases, where there is more of a balance of probability and a lower threshold
- Opportunity by a suspect may be discounted if a plausible and verifiable alibi can be offered to show that the suspect did not have the opportunity to commit the transgression.

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The Motives for Transgressions

- It is not essential to prove motive, and it is often difficult to do so
- However, data may exist on a device that may offer some explanation to possible motivation or, for that matter, an absence of motive and criminal intent.
- Motive may be determined by collecting evidence that links the user to some activities that confirm a degree of knowledge and control over the computer and relevant applications and files used in the transgression
  - Always be wary of the obvious!
- False evidence, too, can relatively easily be generated by mischief- makers out to implicate an innocent party

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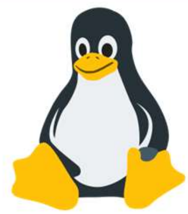
WHERE TO START LOOKING?



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Deciding Where to Look

- If (1) we have a transgression and (2) identified somebody had means, the opportunity, and the motive to commit it using a (computing) device how do we prove the “how and why”? Where should a practitioner start?
- Computers and other devices store information in directory systems of varying sorts, similar to Windows Explorer
    - Knowing common locations and file extensions for files can help speed up the initial investigation → My Documents, Home folder, root, etc.
    - However, the number of files stored on a typical computer makes it impracticable because of time constraints and the fatigue of checking every file
- There are tools that assist in narrowing the search by allowing searches for specific file types, file names, and file locations → both paid commercial/ military grade and open-source versions

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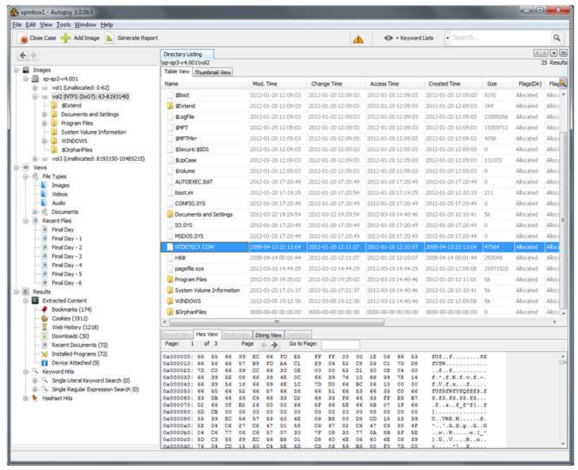
File Category

- With categorization available, practitioners looking for webpage files such as HTML or other categories would find locating and selecting evidence quicker and less tedious
- File categories can be divided into file signature and file types
- File signatures recognize the internal structure and pattern of a file, while file types are based on the application software that uses the files important for files that have been purposefully hidden or obfuscated



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File Indexing and Searching

- Files are often recovered from allocated space, where they are maintained by the operating system in what is called a logical state → easily located and retrieved during forensic recovery
- Files are constantly being created, deleted, updated, etc. across the storage device → the files are left in unallocated data space which is a quick way of the OS marking storage space as available for other data without actually removing/deleting the data
- These remnants of the file remain but will be further eroded and eventually completely overwritten by new files being written to and occupying the same space

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Handling Deleted Files

- Forensic tools allow the practitioner to recover these files or fragments of deleted files that may assist in reconstructing key events in a case
- Data carving is the technique used to undertake the recovery of file fragments and can be done manually using a hex editor or automatically using advanced forensic tools
- Deleted files may be readily recoverable by checking for deleted filenames held in file directories → it is not uncommon for names of files to be reused, files with no filename but with metadata, or filenames and metadata with no content
- Regardless, data recovery can be made much more difficult if a device is password protected and/or encrypted which is fast becoming very popular

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PASSWORD SECURITY, ENCRYPTION & HIDDEN FILES

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User Access to Information

- User access controls offer some degree of protection against unauthorized access
  - Not only is it essential to restrict direct human access to information to those who are authorized, but the information also needs protection from access by other programs, processes, or systems that may be connected to the device
- So why is user access so important?
- User access controls information in terms of its confidentiality, integrity, and availability, which are all at risk and need protection

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Security Triad (CIA) for Digital Information

- **Confidentiality** or privacy is required to prevent unauthorized access to information → a user may be granted and misuse permissions either accidently or covertly
  - This loss of confidentiality means when he/she is investigated there is no record of unauthorized access to their computer and no evidence of any intruder
- Examples
  - Leaving computer unlocked when leaving a room
  - Printing documents to a shared printer
  - Using simple passwords that can be easily "seen"

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Security Triad (CIA) for Digital Information (Cont.)

- **Integrity** refers to the reliability and trustworthiness of an item throughout its lifecycle
  - i.e. information integrity ensures that the information content within a data object remains valid and accurate
- The integrity of information also requires protection: some sort of assurance or guarantee that the information has remained in pristine condition, is unaltered, and is uncontaminated
  - The creator and custodian of the information needs confidence that it has not been altered or corrupted by unauthorized action by human intervention or perhaps by a computer or system glitch
- Information **availability** means that information is accessible to those wishing to use it
  - User may inadvertently deny themselves availability (e.g. accidental deletion), or a system process might render information unavailable (hardware failure) or a malicious party might deny users access (ransomware)

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Storage Encryption

- Encryption has been used to conceal information from unauthorized viewing
  - The goal of encryption is such that even when the message has been intercepted, encryption prevents, or at least hinders, unauthorized viewing of the contents of the message
- Encryption has now moved on to majority of daily devices and applications
- Simple forms of encryption may be cracked through brute-force means whilst advanced encryption may resort to backdoor vulnerabilities
- The problems facing practitioners confronted with locked and encrypted devices is fast becoming a serious challenge

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Summary

What was covered today:

- The variety of computing and storage devices and outlined the nature of digital information they hold of potential evidentiary value
- Introduced and explained the nature of filesystems and outlined some typical files that contain evidence of evidentiary value and where they may be located on devices
- A very small summary of password security and encryption has shown that digital devices are becoming more advanced and are using encryption as an effective security feature

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THANKS – Q&A

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